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Roll number	EP22B020

7.1: Using the gm values from assignment 5.

If the value of C_c used is 1pF as directed by assignment 5, parasitics (mainly the C_{gd} of M_{11} , which is 529.7fF) around C_c are comparable to C_c itself and nearly double its effective value, due to which $\omega_{u,loop}$ becomes nearly half.

This can be compensated for by increasing the gms or by decreasing C_c to bring $\omega_{u,loop}$ back to the expected value.

Here, the external C_c added will be changed to 470.3fF to account for the parasitic capacitance due to the C_{gd} of M_{11} .

Procedure:-

- Slew rate was found using the `slewRate` function of the calculator in cadence.
- The swing limits were taken as the input DC voltages for which the output was found to not remain linear and saturate to the supply rails.
- HD_3 was found from the ratio of the amplitudes of the 3rd harmonic and the fundamental component from the dft option of the calculator in cadence.

Pictures of plots and testbenches/schematics used have been attached at the end.

a)

Specifications and component value

Specification	Value	Unit
Closed loop dc gain k	8	-
Closed loop bandwidth f_B	20	MHz
Load capacitor C_L	24	pF
Load resistor R_L	663.14	Ω
Input resistance R_i	10	k Ω
C_c	470.3	fF
Component parameter	Value	Unit
L (all)	0.3	μm
W (all)	1	μm
m_{00}	2	-
I_{00}	79	μA
$V_{\text{dsat}00}$	251.6	mV
m_0	7	-
$V_{\text{dsat}0}$	249.4	mV
$m_1=m_2=m_5=m_6$	12	-
$V_{\text{dsat}1}$	149.3	mV
$V_{\text{dsat}2}$	149.4	mV
$V_{\text{dsat}5}$	143.8	mV
$V_{\text{dsat}6}$	144.6	mV
VB56	1.13	V
$V_{\text{DS}1}$	199.9	mV
$V_{\text{DS}2}$	198.3	mV
$m_3=m_4=m_7=m_8$	45	-
$V_{\text{dsat}3}$	-156.4	mV
$V_{\text{dsat}4}$	-156.4	mV
$V_{\text{dsat}7}$	-156.6	mV

Vdsat ₈	-155.3	mV
VB78	0.93	V
VDS ₃	-205.2	mV
VDS ₄	-206.9	mV
m ₁₂	166	-
Vdsat ₁₂	252	mV
m ₁₁	850	-
Vdsat ₁₁	-249.9	mV

Transconductances

Transconductance	Value	Unit
Gm1 (assignment 5)	1.13	mS
gm ₁	1.141	mS
gm ₂	1.14	mS
Gm ₂ (assignment 5)	40.87	mS
gm ₁₁	41.5	mS

b)

Result	Value
Closed loop DC gain	18.0503 dB = 7.989
Closed loop 3-dB bandwidth	20.658 MHz
Unity loop gain frequency	15.125 MHz
Phase margin	76.732°
rms output noise	0.796 mV
Opamp open loop DC gain	57.526 dB = 752.14
Positive slew rate	208.8 MV/s
Negative slew rate	135.7 MV/s
Positive swing limit	100 mV
Negative swing limit	-100 mV

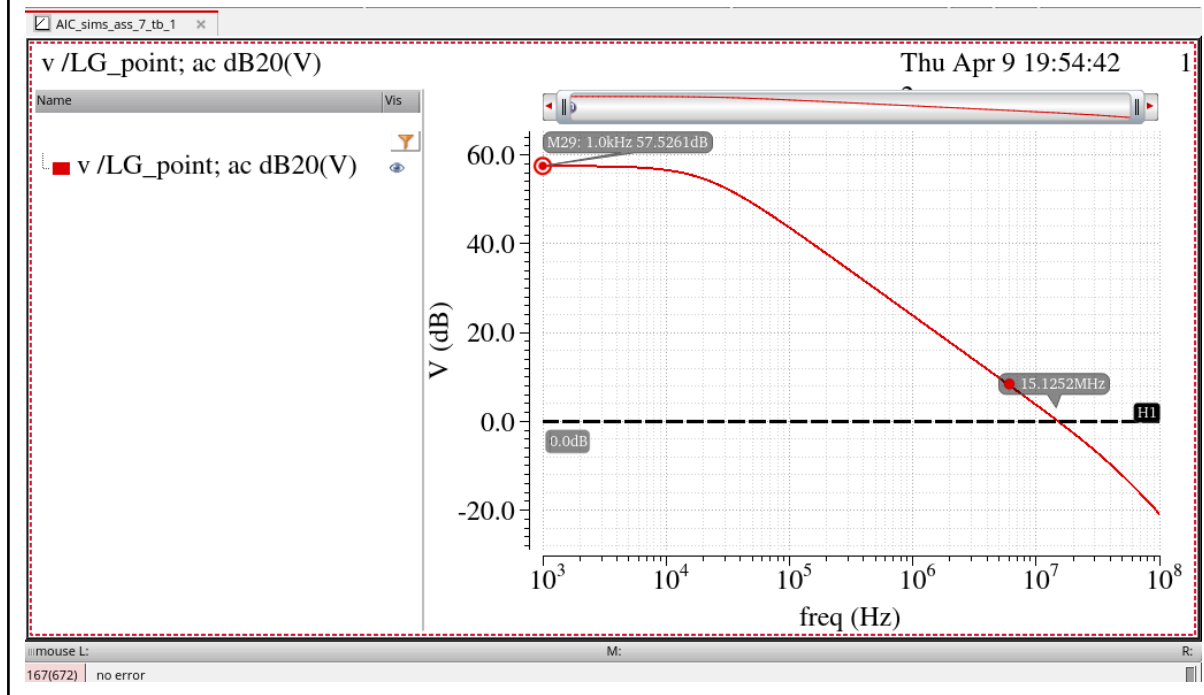
HD ₃	0.0392 = -28.13 dB
Supply voltage	1.8 V
Current consumption	6.918 mA

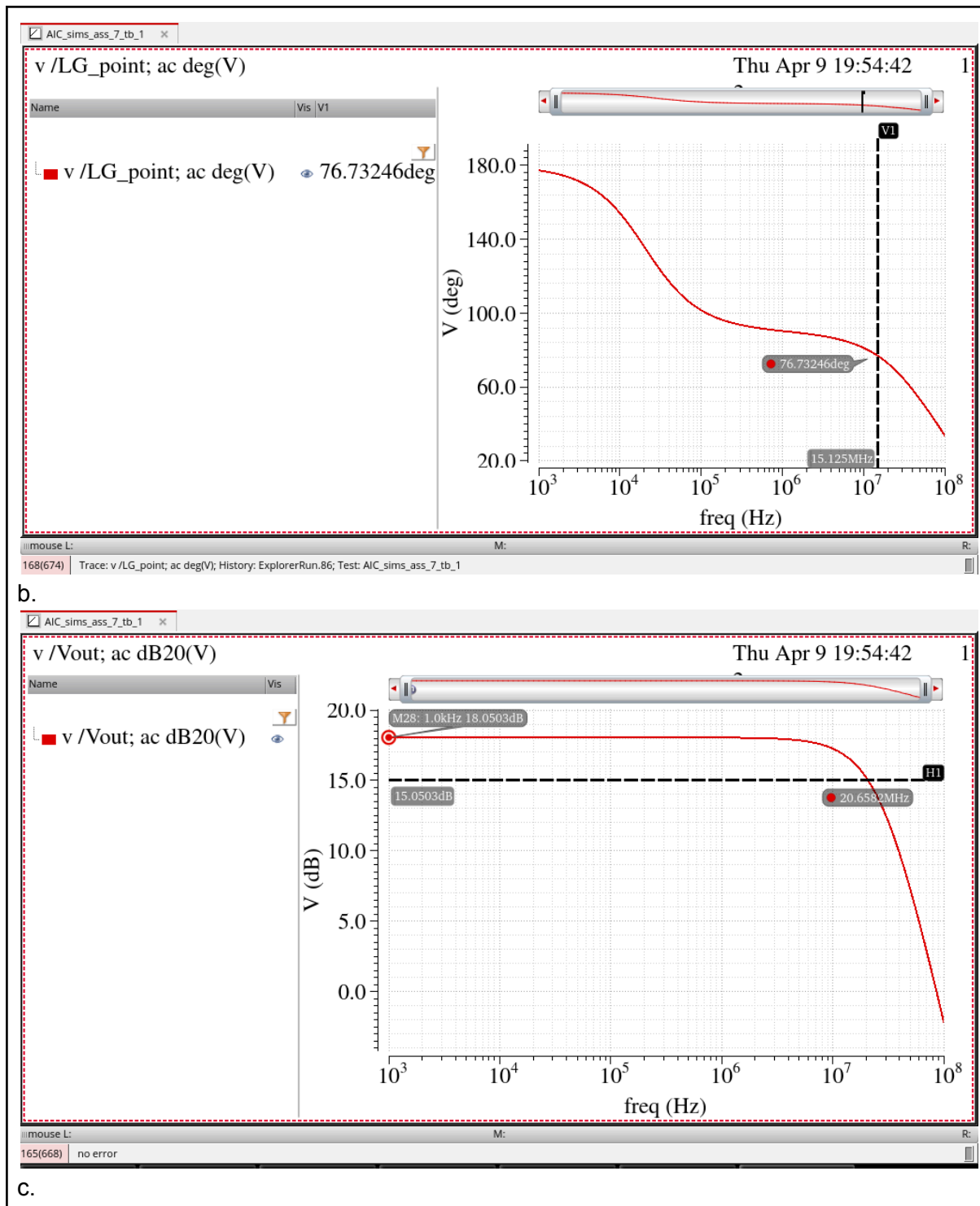
Noise variance contributions

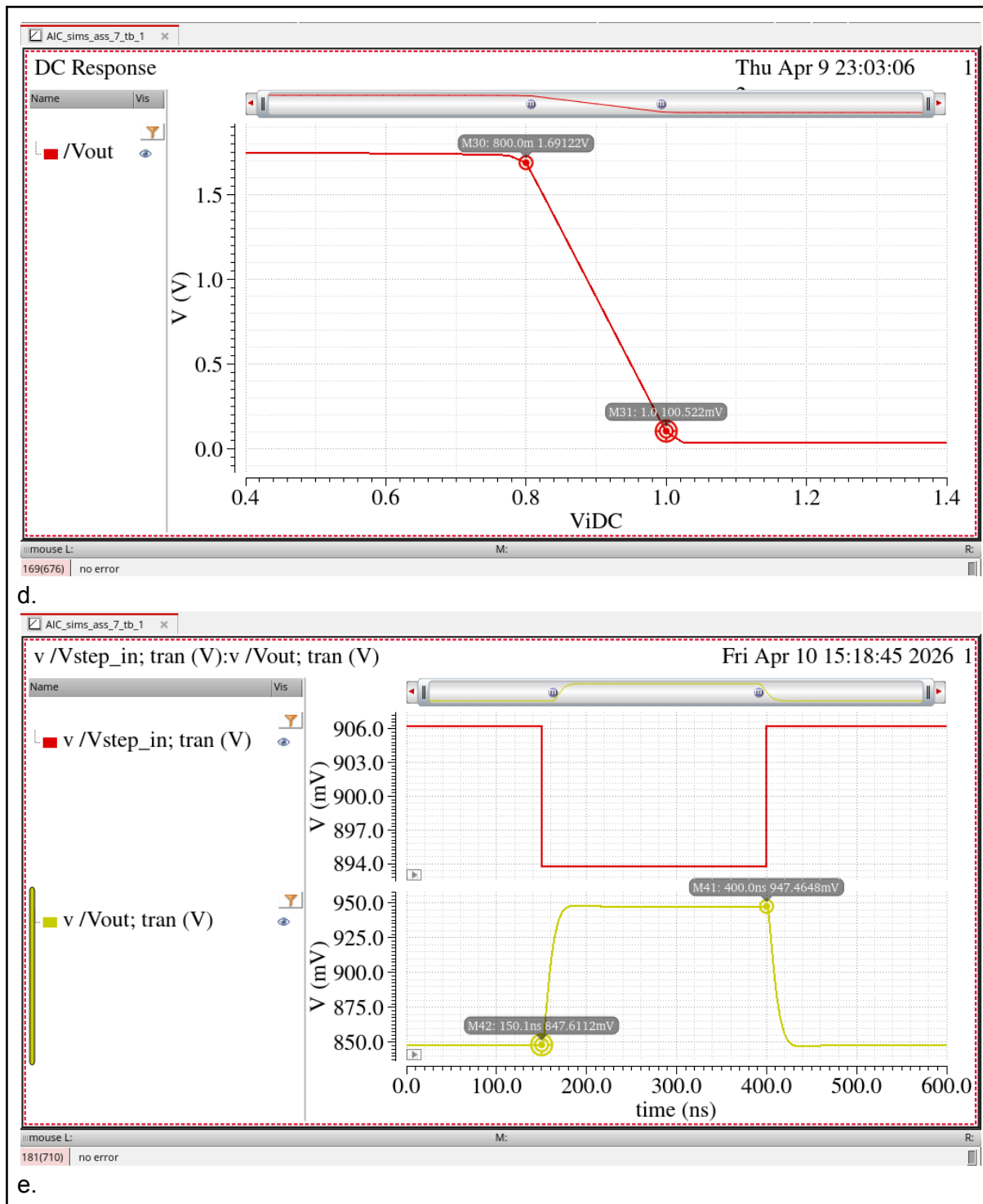
Component	Percentage
Ri	51.79
Rf	6.47
First stage (M ₁ -M ₈)	39.25
Second stage (M ₁₁)	0.02
RL	0.00

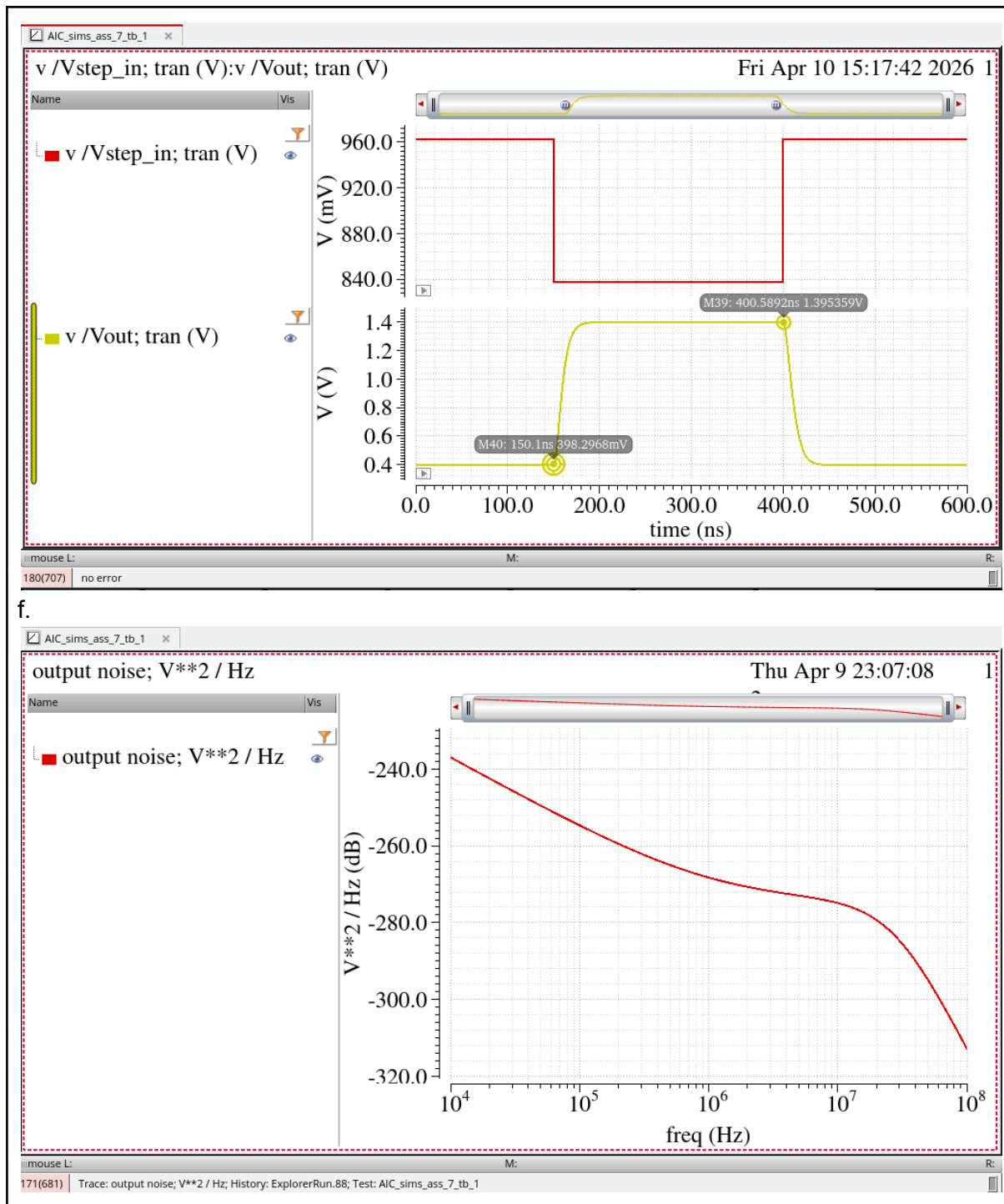
c) Plots:-

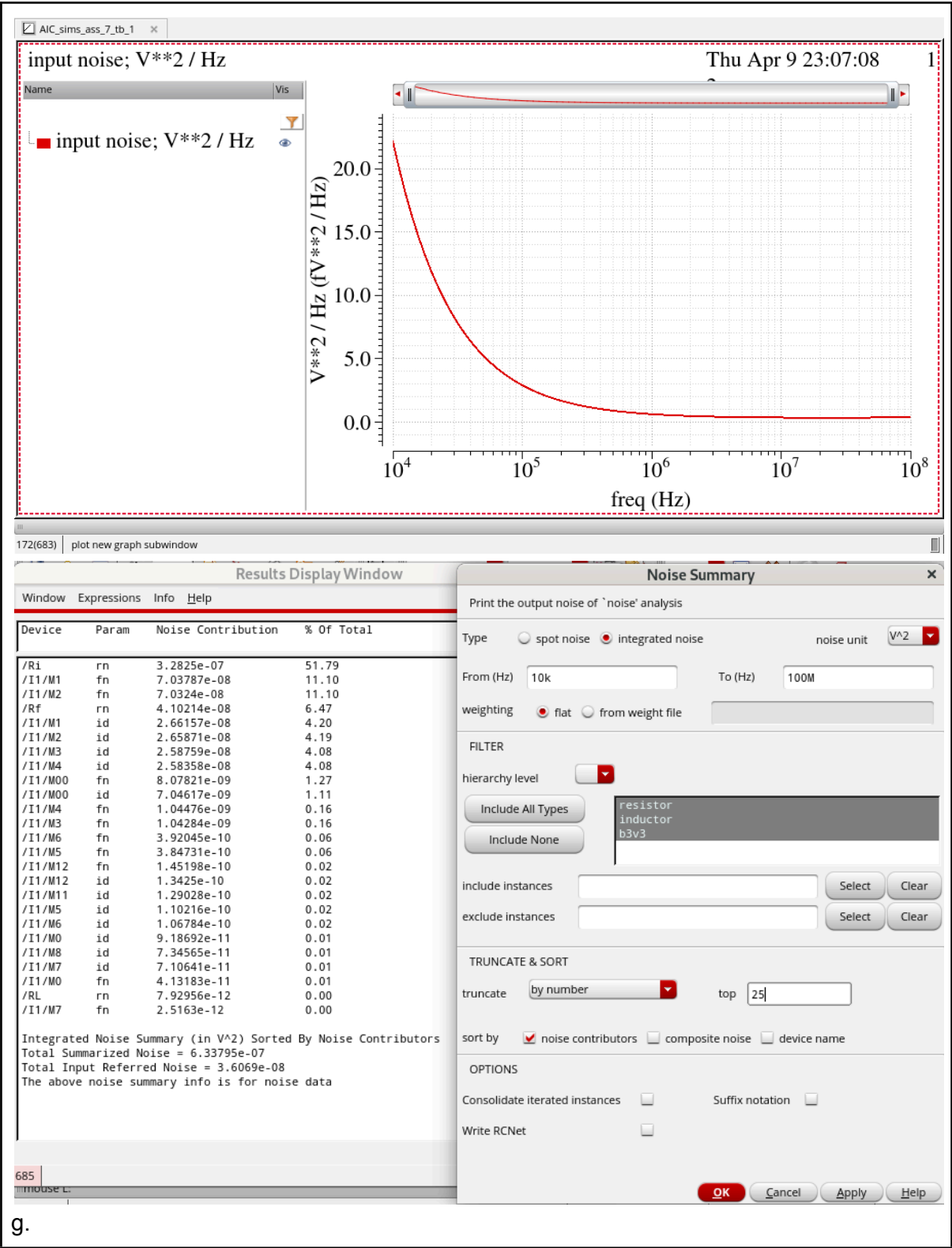
a.



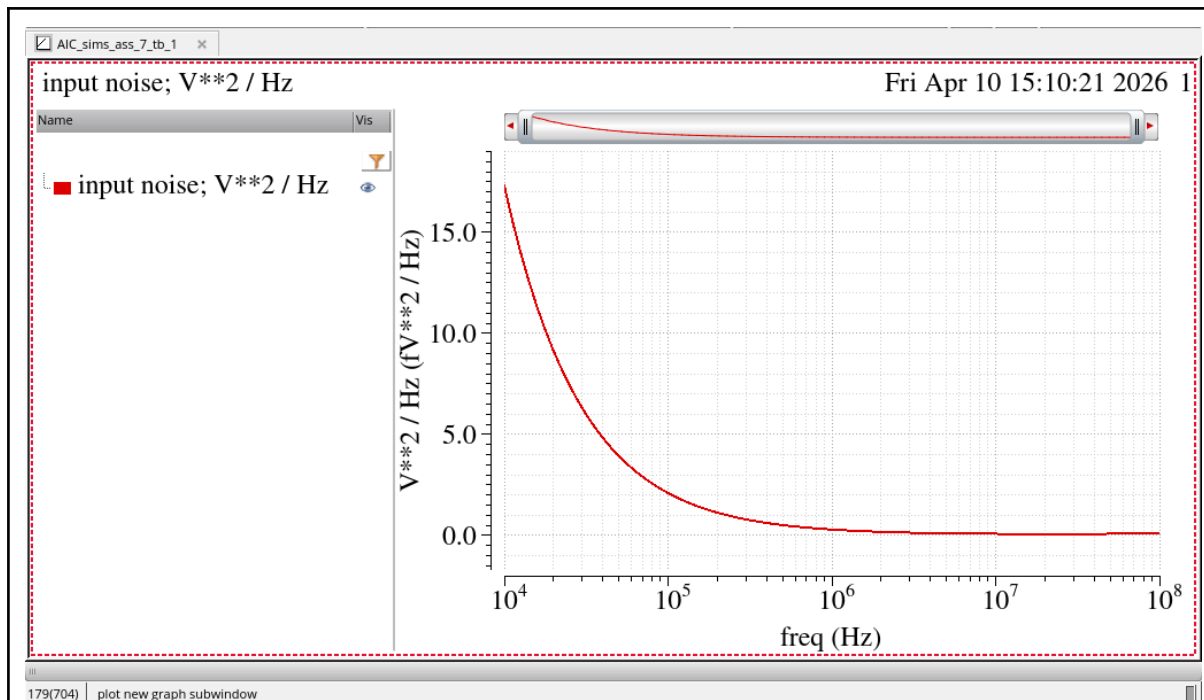






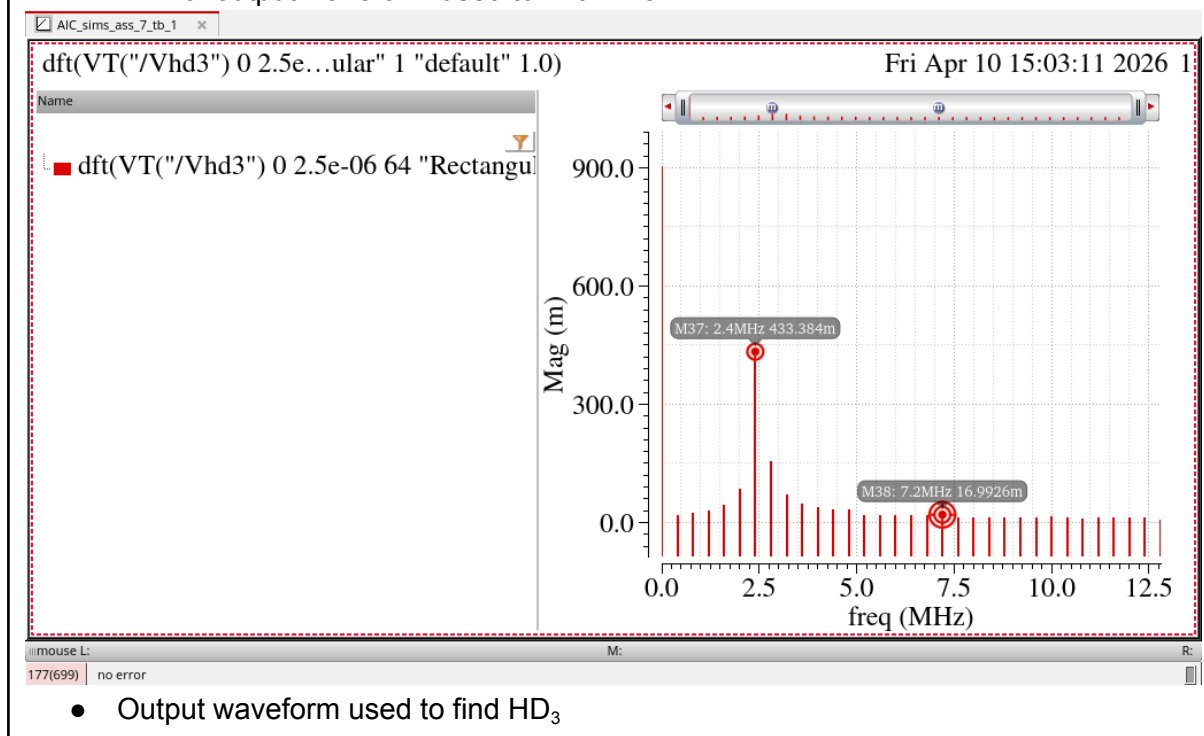


g.

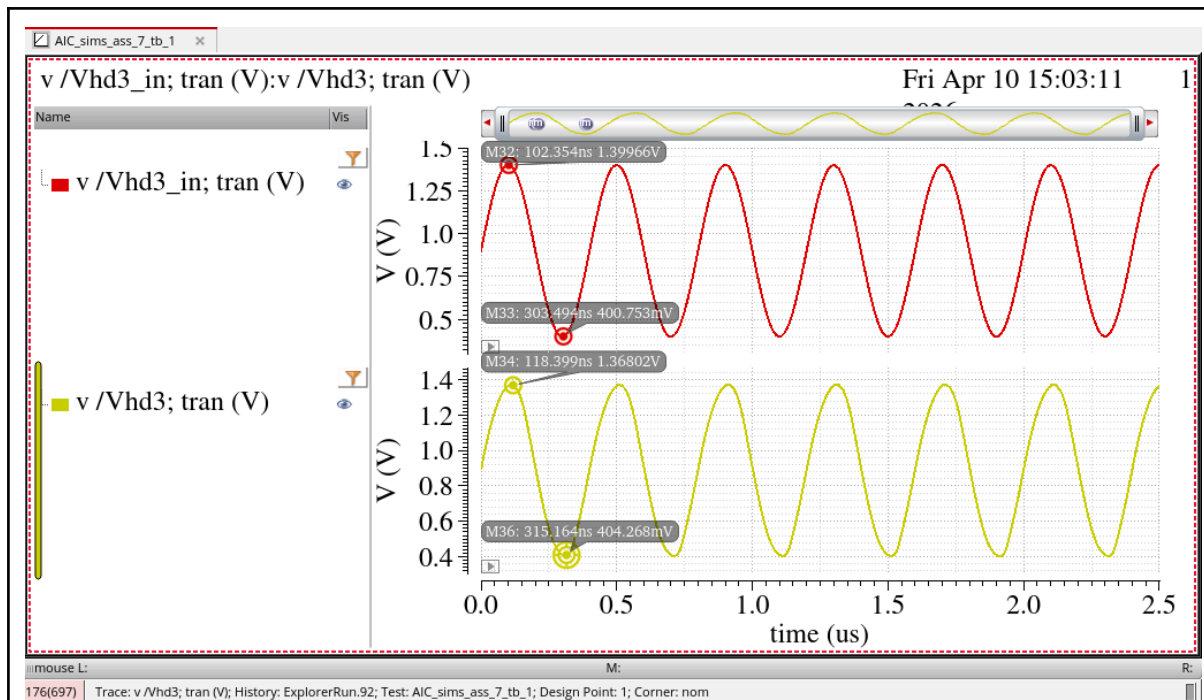


Miscellaneous

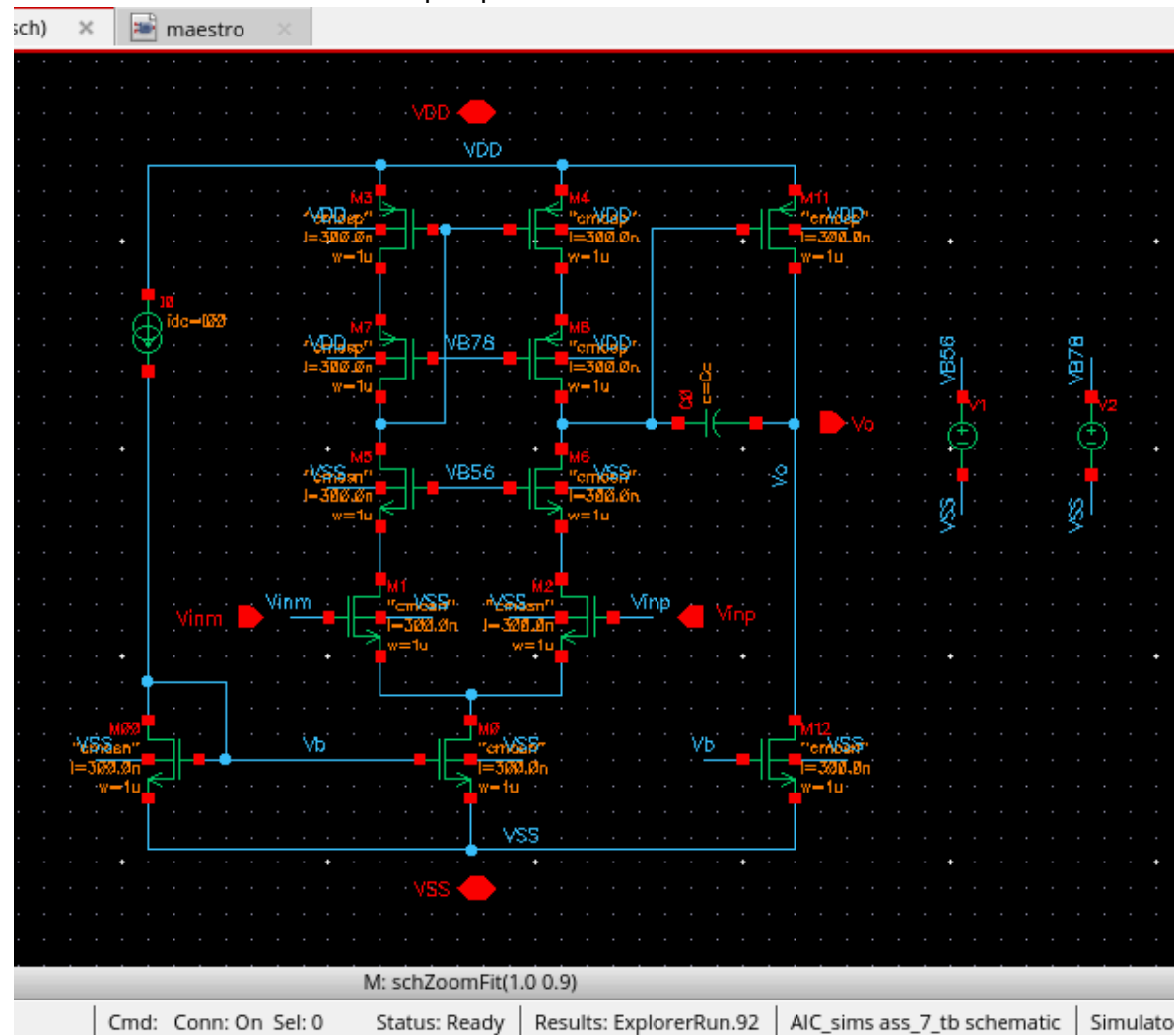
- DFT of output waveform used to find HD₃:-



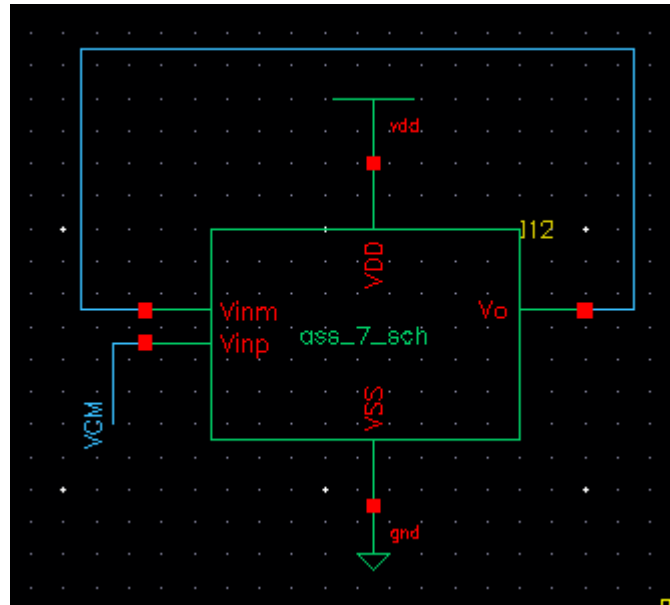
- Output waveform used to find HD₃



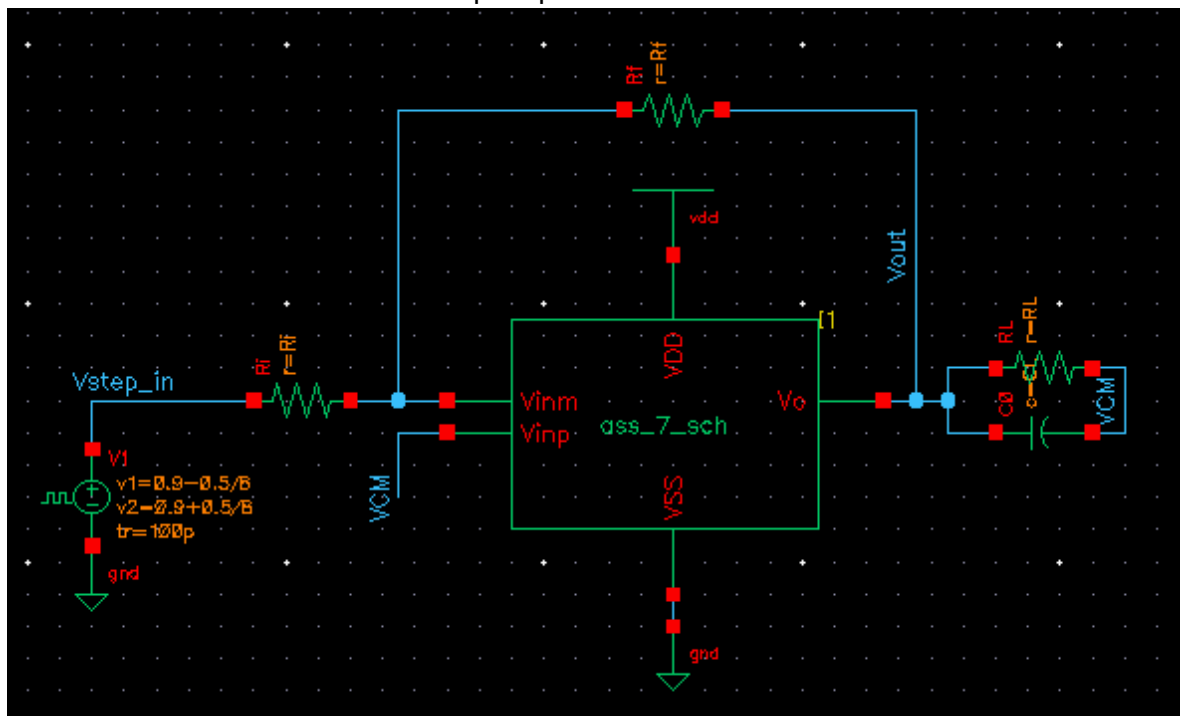
- Schematic used for the opamp



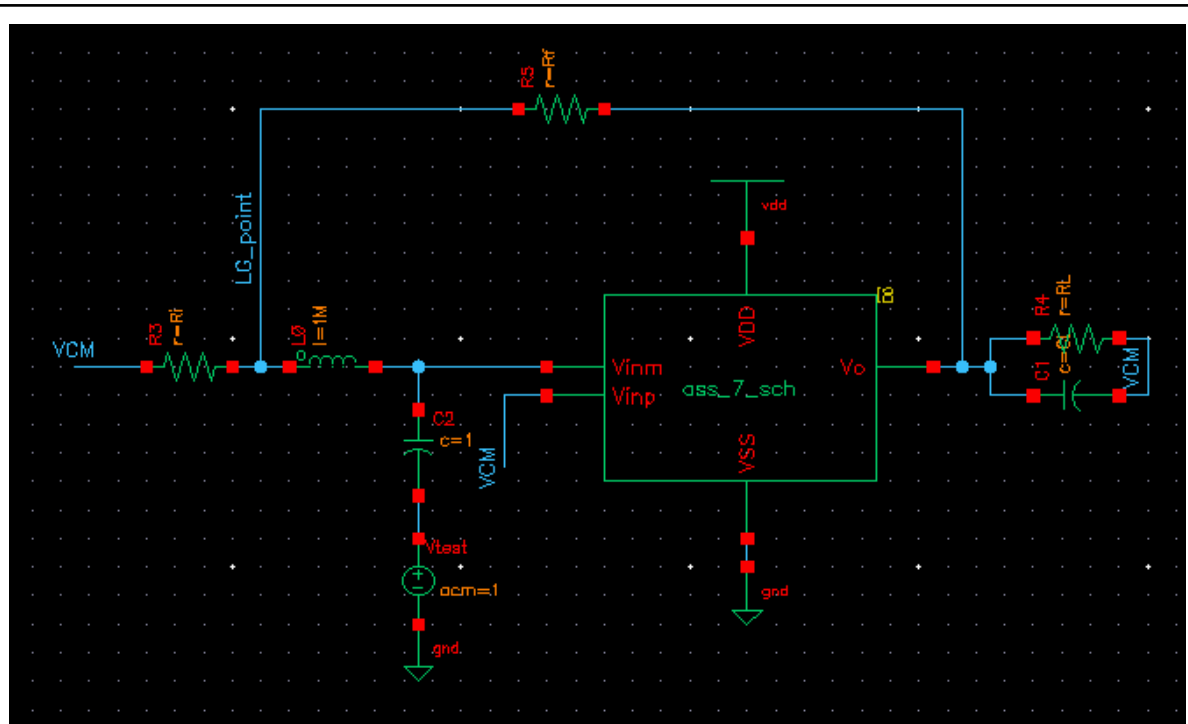
- Testbench used for DC simulations



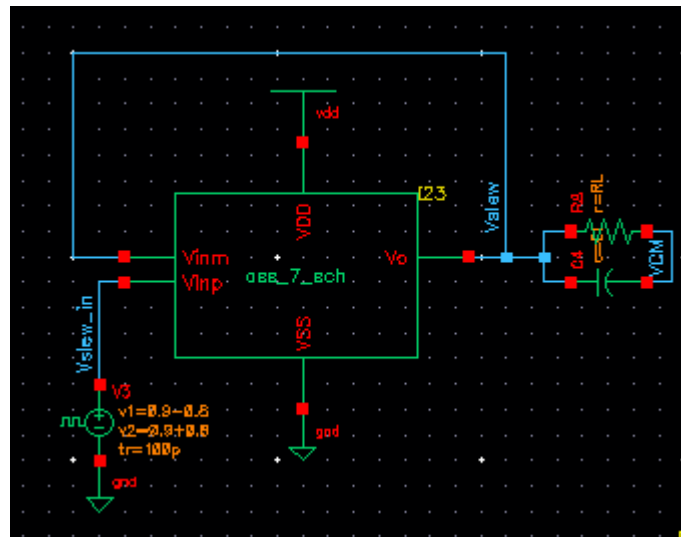
- Testbench used for CLTF/step response simulations



- Testbench used for loop gain simulations



- Testbench used for slew rate simulations



- Testbench used for HD3 simulations

